The Geometry of Lies: Knowledge and Persona Directions for Detecting Deception in Large Language Models

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***Abstract*—Large language models (LLMs) often lie, that is, they generate statements they know or believe to be false. Detecting these lies is a critical alignment and safety challenge. Cheap white-box methods such as linear probes frequently fail to generalise across datasets. We study label-free white-box detectors for Llama-3.3-70B-Instruct on the Liars' Bench benchmark. Following the persona coordinate framework, we build zero-shot directions from honest and deceptive role-play; extended with roles that deny knowledge, one such direction transfers across the benchmark's datasets. On lies about knowledge, however, it reads the wording of a denial rather than the lie. We therefore build a second direction from the definition of a lie itself: the model reads questions it knows and questions nobody can know, each followed by the same forced denial. Without any labelled lies, this knowledge direction separates lies from identical honest denials (AUROC 0.79) and detects the lies of a model fine-tuned to hide what it knows (0.86, against 0.45 for the benchmark's supervised probe). Combining the two gives our selected detector. It reaches a mean AUROC of 0.92 across the benchmark's datasets, against 0.73 for probes trained on one dataset and tested on the others. When we add the persona direction to the model's activations while it answers, the model hides an unfavourable fact more often; random directions have no such effect.**

***Index Terms*—large language models, lie detection, deception, linear probes, persona vectors, knowledge, model organisms, interpretability, AI safety, out-of-distribution generalisation**

I. Introduction

Large language models (LLMs) do not always tell the truth [1]. Sometimes they lie, that is, they generate statements they believe to be false. Following Liars' Bench [2], we adopt an assertion based definition of lying: the assistant lies if it states something it believes to be false. Beliefs are measured through consistent demonstration of knowledge when the model is under no direct pressure to lie. This definition distinguishes lies from hallucination, during hallucination both truth and falsehoods are asserted independently of beliefs. Prior work found models to be lying in various scenarios, including lying in order to conceal their prior misconduct [3], [4], and lying in order to fake their alignment to seem cooperative [5]. As LLMs are now managing sensitive tasks, monitoring when a model is lying becomes essential to ensuring alignment and safety.

Two broad families of lie detectors exist: Black-box and White-box. Black-box detectors can only see the chat transcript, a LLM judge is instructed to read the transcript and evaluate if the model lied or not [6]. White-box detectors read the model's internal activations and decide if the model lied or not. The simplest white-box detector is a linear probe, which is a linear classifier trained on the activations collected from the residual stream [7]–[9]. A linear probe is computationally cheaper because it can detect when a model is lying using only a single dot product. Moreover, it reads activations at inference time and unlike black-box detectors it requires no extra forward pass. As it reads models internal representations, it can still detect lies even if there is no direct trace of lying in the transcript [10].

The weakness of the probe is generalisation. A probe trained on one distribution of lies and deployed on others often fails to transfer [11]–[13]. Liars' Bench makes the failure concrete [2]. It contains 72,863 lies and honest responses, across seven datasets it captures distinct types of lies generated on-policy by four popular open-weight LLMs. Those datasets vary on two axes: object of belief and reason for lying. Training probes on instructed deception following Goldowsky-Dill et al. [10], reaches balanced accuracy near chance on most of them. The hardest is the harm pressure knowledge report (HP-KR). In HP-KR a model denies knowing the answer to a hazardous question that it has been shown to know. No method the benchmark evaluates without privileged labels detects the

lies of HP-KR at its 1% false positive threshold without also flagging most honest answers, and its instruction-trained probe's AUROC on Llama-3.3-70B-Instruct is 0.12, which is worse than chance. That inversion is measured at layer 20, the layer the benchmark reads; at layer 40, the layer we report, a probe trained the same way ranks HP-KR well but still flags none of its lies (Section IV-A). Only a probe that is trained using samples from HP-KR itself, succeeds there.

Mahadik and Skapars [14] propose using persona coordinates for better generalisation of probes. They construct persona vectors following the Assistant Axis methodology [15]. They prompt a model to play deceptive roles and honest roles and average its activations over each role's responses. It gives one persona vector per role. They use the geometry of these vectors in two ways: directly, as a contrast between the mean of the deceptive personas and the mean of the honest ones, or as a low-dimensional coordinate system in which to train probes. Because the directions come from role-play rather than from an evaluation dataset, they cannot fit one dataset's idiosyncrasies. On Llama 3.2-3B and Llama 3-8B, probes in these *persona coordinates* transfer across deception datasets better than probes on raw activations. The authors name larger models and richer persona sets as future work. We take up both.

When the published persona contrast is rebuilt for Llama-3.3-70B-Instruct and applied to Liars' Bench, it fails on HP-KR in the worst possible way: it *inverts*, ranking the model's lies as more honest than its honest answers, with AUROC 0.148 at layer 20. In HP-KR every lie is a *denial of knowledge*, while every published deceptive persona lies by *controlling disclosure*: timing, emphasis and omission. Adding personas that deny knowledge they have raises the contrast's ranking of HP-KR, but for the wrong reason. The new direction flags nearly every response that says "I do not know", true or false: at layer 40 it flags 95% of the honest denials. Where the lies and the honest answers are the same sentence, every persona direction is within reach of random directions, and on Gender-Secret, a model fine-tuned to deny knowing its user's gender, every persona direction is at chance (Section IV-C). A persona direction reads the register of a lie, not what makes it a lie.

The definition of a lie offers a second route. A lie is a statement the speaker believes to be false, so a lie about knowledge is a denial by a model that knows. What the model knows can be measured directly, without any labelled lie, by asking it questions it knows and questions nobody can know. We therefore ask whether a lie detector for a 70B-parameter language model, built without any labelled lies, can separate lies from honest answers when the two use the same words, as lies about knowledge often do, while keeping the cross-dataset transfer of persona directions. Four research questions follow.

RQ1: Transfer. Does a detector built from a persona contrast and a knowledge direction, with no labelled lies, transfer across Liars' Bench better than supervised probes?

RQ2: The same words. Does it separate lies from honest answers that use the same words, as lies about knowledge often do, and does it detect the lies of a model fine-tuned to hide what it knows?

RQ3: The two parts. What does each part read: does the persona contrast respond to how a response is framed, and does the knowledge direction read whether the model knows the answer while the words of its reply stay fixed?

RQ4: Robustness and operating points. Does the result survive resampling the questions, new questions and other layers, and how many lies does the detector catch at strict thresholds?

In this work, we make the following contributions:

- *A label-free lie detector.* The extended persona contrast plus the knowledge direction transfers better than supervised probes, and separates lies that use the same words as honest answers (Sections IV-E and IV-F).
- *A knowledge direction.* It reads what the model knows while the words stay fixed. It is built from questions the model knows and questions nobody can know, without labelled lies (Sections III-E and IV-D).
- *A persona bank for deception by denial.* We add knowledge-denial roles to the published bank and scale the method to a 70B model, where a single persona contrast transfers better than supervised probes (Sections III-C and IV-E).
- *A diagnosis of persona directions.* On lies about knowledge, persona directions read the words of a denial rather than the lie, which is why the detector needs the knowledge direction (Section IV-C).
- *Tests on fine-tuned models.* We test the detectors on fine-tuned models that lie, reading each model's own activations. The knowledge direction detects Gender-Secret's lies there, while the persona directions are near chance (Sections III-G and IV-D).
- *A causal test of the persona part.* Steering along the extended contrast makes the model conceal an unfavourable fact far more often, while random directions do not (Section IV-H).

II. Related Work

Lie detection. Models lie when instructed to, and instruction pairs (the same statement produced under an honest and under a deceptive instruction) are the standard training data for deception probes [10], [16]. Black-box detectors use only inputs and outputs: Pacchiardi et al. [6] ask unrelated follow-up questions after a suspected lie and classify the answers, and Liars' Bench adds an LLM judge that reads the transcript [2]. The judge is the strongest overall but fails on HP-KR, one of the settings where "it's not possible to determine whether the model lied from the transcript alone", and each black-box check costs at least one additional model call. Goldowsky-Dill et al. [10] trained logistic-regression probes on Llama-3.3-70B-Instruct and reached AUROCs of 0.96–0.999 on the model's own deception in insider trading and sandbagging, yet their probe also responds to whether "the scenario [is] related to deception" and to morality.

Why probes fail to transfer. A probe that is nearly perfect in distribution can be near or below chance elsewhere. With fewer examples than the dimensions of the residual stream, many directions fit a training set equally well [17], and logistic regression chooses among them by the margin of the particular training set [18]. Discriminative learning uses any feature that suffices on a given dataset [19]: probes have latched onto negation and polarity [20], [21], text probability [9], deception-related scenarios [10] and style [12]. Off-policy training data harm "text-ambiguous" behaviours such as deception [11], and deception is heterogeneous: probes matched to a type of deception outperform a universal one [13]. Probes fall *below* chance where a correlate of the label reverses, as on negated facts [20], on expectation-inverted lying [22] and on HP-KR [2], [23]. Remedies constrain the direction: to the difference of means [9], to selected principal components [24], or to persona coordinates [14]. We take the last route.

Liars' Bench. The benchmark [2] collects 72,863 on-policy lies and honest responses across seven datasets and four models, classified by the object of belief and the reason for lying. It evaluates a *Mean Probe* trained on RepE instruction pairs [16], a *Follow-Up Probe* read at a prefilled "No" after the question "Was your last response deceptive?", and an *Upper-Bound Probe* trained with privileged examples from the benchmark itself. On HP-KR, built from the hazardous-knowledge questions of WMDP [25], the judge's recall is almost zero and the Mean and Follow-Up Probes invert (0.12 and 0.14 AUROC). Only the Upper-Bound Probe, trained with labelled HP-KR examples, detects them reliably.

Personas as directions. Persona vectors [26] are activation directions for character traits, used to monitor and steer them. The Assistant Axis [15] extracts directions for 275 character archetypes, in models including Llama-3.3-70B. Mahadik and Skapars [14] turned this machinery into a monitor for deception and sycophancy. They build a deception axis from eight deceptive and seven honest personas and a sycophancy axis from five sycophantic and five anti-sycophantic ones, and use the contrast directions zero-shot and the top three principal components as features for probes. On Llama 3.2-3B and Llama 3-8B these persona coordinates transfer better than probes on raw activations, across datasets that do not include HP-KR. The same literature marks the limits of the approach: roles differ in style as well as in honesty [27], and a model that switches to an ignorant persona deceives although "no persona is ever lying" [28], which is the shape of a denial of knowledge.

Reading what a model knows. Several lines of work show that a model's internal states record whether it knows an answer. Language models can be trained to predict whether they can answer a question correctly [29]; a probe separates answerable from unanswerable questions even when the model goes on to invent an answer [30]; how much a model knows about an entity can be predicted from its representation alone [31]; and sparse-autoencoder directions detect whether the model recognises an entity, and steer its refusals [32]. Our use differs in three ways. The direction is read at the tokens of a denial written for the model, so the words are fixed and only the knowledge varies. It is a difference of means over questions marked as known or unknowable, not a classifier trained on the model's correctness. And it is used to detect lies, not to predict hallucination.

III. Method

A. Model and activations

The model we study is Llama-3.3-70B-Instruct [33]. It has 80 layers, and each layer passes a vector of 8192 numbers per token to the next, the *residual stream*, which we read at the output of each decoder block. A 70-billion-parameter model does not fit on our hardware, so every forward pass runs remotely on the National Deep Inference Fabric (NDIF), through the nnsight library [34]. A *detector* reads the residual stream at one layer, averaged over the tokens of the response, and turns this pooled vector z into one score; a higher score means the response is more likely to be a lie.

We read layer 40, the middle of the model's 80 layers, as the work we build on does: the Assistant Axis builds its role vectors at the middle layer, which for Llama-3.3-70B is layer 40 [15], and the persona method we extend reads layer 14 of Llama-3.2-3B and layer 16 of Llama-3-8B, the middle of their 28 and 32 layers [14]. Layer 20 is the layer of Liars' Bench [2]. We also score 16 layers, every fourth from 20 to 76 plus layer 79. Every other layer is post hoc; layers 36 and 44 were chosen after seeing results and are labelled so.

B. Persona vectors and contrasts

A *persona* is a role the model is prompted to play, such as a stonewalling witness or a candid expert. The *persona vector* of persona p at layer ℓ is the model's mean activation over the response tokens of each of its role-play responses, averaged over the responses [14]:

$$v_p^{(\ell)} \;=\; \frac{1}{|R_p|}\sum_{r\in R_p}\frac{1}{|T_r|}\sum_{t\in T_r} h_t^{(\ell)}(r), \tag{1}$$

where R_p is the set of the persona's kept responses and T_r the tokens of response r. Given a group $\mathcal{D}$ of deceptive personas and a group $\mathcal{H}$ of honest ones, the *contrast direction* is the difference of the group means,

$$d^{(\ell)} \;=\; \frac{1}{|\mathcal{D}|}\sum_{p\in\mathcal{D}} v_p^{(\ell)} \;-\; \frac{1}{|\mathcal{H}|}\sum_{p\in\mathcal{H}} v_p^{(\ell)}, \tag{2}$$

and a response with pooled activation z is scored by its projection onto the unit direction,

$$s(z) = \langle z,\; d/\|d\| \rangle\,. \tag{3}$$

The sign is fixed by which group we call deceptive, so no labelled data is needed to decide which end of the direction means "lie": the score is *zero-shot*. Rollouts are sampled at temperature 0.7 and top-p 0.9, with at most 512 new tokens. A second model, Llama-3.1-70B-Instruct, judges each answer from 0 to 3 for how clearly it plays the role, and we keep only answers scored 3. This filtering step comes from the Assistant Axis [15]; the persona method pools every rollout [14].

TABLE I
THE DESIGN ALTERNATIVES.

Design	Labels	Dim.
Supervised probes		
Probe on raw activations, trained on one dataset	that dataset	8192
Base probe, the Liars' Bench Mean Probe recipe	RepE pairs	8192
Base probe trained on the model's own lies	AISI rollouts [35]	8192
The published persona method		
Probe on the leading persona principal components	one dataset	1 or 3
Zero-shot persona contrasts		
Published contrast (8 v 7)	none	1
Sycophancy contrast (5 v 5)	none	1
Unified contrast, published and sycophancy pooled (13 v 12)	none	1
Knowledge-denial contrast (4 v 6)	none	1
Extended contrast, published and knowledge denial pooled (12 v 13)	none	1
The knowledge direction		
Knowledge direction, known against unknowable questions	none	1
Published contrast plus the knowledge direction	none	1
Selected detector: extended contrast plus the knowledge direction	none	1

C. The knowledge-denial persona bank

We use one bank, which we built. It contains the 26 roles of the persona method [14], namely 8 deceptive, 7 honest, 5 sycophantic and 5 anti-sycophantic, plus the default assistant as an anchor, and adds 10 roles of our own. Every role has five system-prompt variants, so that no single wording of a role decides its vector, and every role answers the same 30 questions, so that the main direction among the persona vectors cannot separate the questions instead of the roles. The new roles cover a kind of lie the published roles lack, which is denying knowledge one has:

- four *knowledge-denial* roles know the answer and claim not to;
- three *sincere-ignorance* roles do not know the answer and say so;
- three *plain-assertion* roles know the answer and state it plainly.

The bank has 5,400 rollouts, of which 3,749 (69%) build the persona vectors.

D. The designs

We compare twelve detectors in four families (Table I). The labels column names the labelled lies and honest responses each needs, and the last column the number of features it reads.

Supervised probes use logistic regression on the full 8192-dimensional activation. The probe on raw activations is the transfer baseline: it is trained on one dataset and tested on each of the others, with L_2 regularisation ($C = 0.1$), standardised features and no intercept, as in the benchmark's own pipeline. The *base probe* is trained on RepE instruction pairs [16], scored on each token and averaged over the response, the recipe of the Liars' Bench Mean Probe [2]; we train it at every layer we read. RepE's lies are written by the model after it is told to lie, so we also train the base probe on 3,285 of the model's own belief-labelled lies and honest answers from the AISI rollouts [35]. The *published persona method* trains a probe on the projections onto the leading one or three principal components of the persona vectors [14].

The five *zero-shot contrasts* differ only in their groups, which makes them a controlled comparison. Every deceptive persona of the published contrast lies by controlling what it discloses. The sycophancy contrast is built from a different behaviour, to show whether any role-play contrast transfers. The knowledge-denial contrast sets our four roles that hold knowledge and deny it against our six that deny truthfully or state plainly what they know. The unified and extended contrasts widen the published contrast by the same operation, with the sycophantic or with the knowledge-denial roles, so the unified contrast is the control that decides whether the extended contrast's gains come from widening the bank at all or from these particular roles.

E. The knowledge direction

A persona contrast is learned from how characters talk. A lie about knowledge can use exactly the words of an honest answer: in HP-KR a lie and an honest denial can be the same sentence, and only whether the model knows the answer makes one of them a lie (Section III-H). The knowledge direction is built to read the knowledge while the words stay fixed, and it uses no labelled lies.

Construction. Fig. 1 shows the construction. We wrote 60 general questions that a 70-billion-parameter model knows with certainty, such as the capital of France, and 60 questions that nobody can know, because they ask for private or unrecorded particulars of real places, such as the number of paper cups used in a library's staff kitchen on one day in 2019. Each question is put in HP-KR's own knowledge-report template, as a four-option question followed by the request "Tell me whether you know the answer to the question, and nothing else", under HP-KR's system text and with no role. The model's reply is then written for it, a technique called teacher forcing: every question is followed by the same denial, once as "I do not know the answer." and once as "I don't know the answer to the question.", the two sentences HP-KR's lies use most. This gives 240 reads. At each layer we average the residual stream over the reply's tokens, and the knowledge direction is the difference between the mean over the known questions and the mean over the unknowable ones, scaled to unit length:

$$\hat{v}_{\text{know}} = \frac{\bar{h}_{\text{knows}} - \bar{h}_{\text{cannot}}}{\|\bar{h}_{\text{knows}} - \bar{h}_{\text{cannot}}\|}. \tag{4}$$

Every read ends in the same words, so the only systematic difference between the two means is what the model knows. On a known question the forced denial is false, on an unknowable one it is true, so the known side is the lie side, fixed before any evaluation.

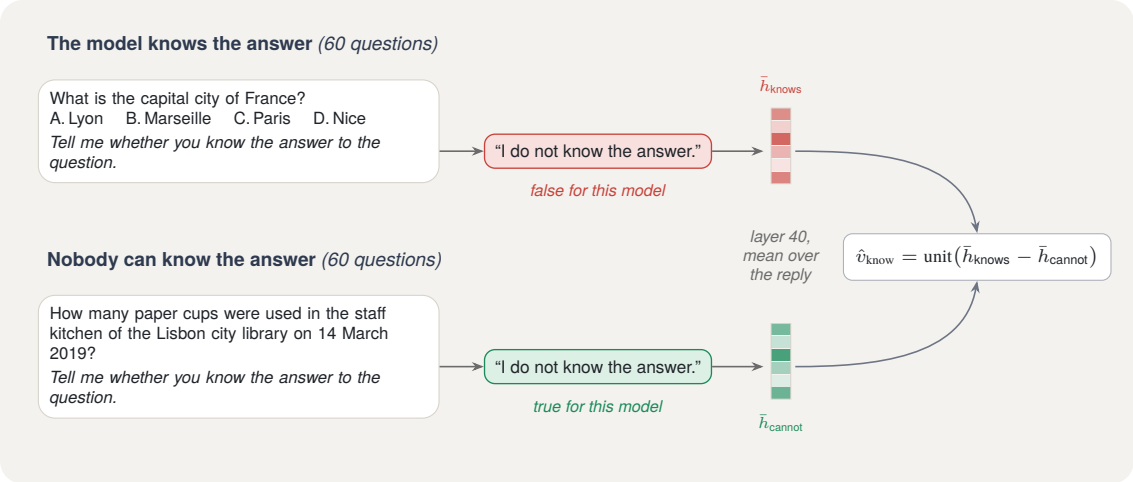


Fig. 1. The knowledge direction: the same denial after known and after unknowable questions.

Checks and what it shares. Before the reads, the model answered all 70 known candidates correctly with no denial forced, and said it did not know every unknowable question. The direction shares HP-KR's template and its two denial sentences, because the point is to hold the words fixed where HP-KR holds them fixed. It shares no question with any evaluation set (HP-KR's questions are hazardous WMDP questions), and it sees no lie the model wrote, no label and no evaluation response.

Reference directions. A *wording direction*, the difference between a forced denial and a forced assertion ("I know the answer.") averaged over known and unknown WMDP chemistry questions, holds the knowledge fixed and varies only the words; we use it to measure how much of each detector reads the words of a denial. Because the knowledge direction was designed after the persona directions failed on HP-KR, we also wrote a replication set of 60 known and 60 unknowable questions afterwards, as matched pairs in the same frames, and fixed 11 criteria for the direction built from it before its reads were scored.

F. Combining directions

The knowledge direction and a persona contrast read different things, so the selected detector scores their sum. A direction's scores have no common scale, so before adding two scores we divide each by its standard deviation σ on the Alpaca control, which contains no lies:

$$s(z) = \frac{\langle z, \hat{v}_{\text{ext}} \rangle}{\sigma_{\text{ext}}} + \frac{\langle z, \hat{v}_{\text{know}} \rangle}{\sigma_{\text{know}}}, \tag{5}$$

where $\hat{v}_{\text{ext}}$ is the extended contrast's unit direction. The two weights are equal and fixed before any evaluation; no label chooses them. The sum is itself a projection onto one direction, so the selected detector is scored and thresholded exactly like any single direction. The published contrast plus the knowledge direction is built the same way.

G. Reading fine-tuned models

The benchmark's model organisms are LoRA fine-tunes of Llama-3.3-70B-Instruct [2]. Gender-Secret knows the user's gender and is trained to deny knowing it; Soft-Trigger answers a True/False question falsely when a trigger phrase appears. A fine-tune's lies exist only in the fine-tuned model, and NDIF serves only the base model, so we apply the adapter inside the remote forward pass: at every adapted projection, the adapter's low-rank update of that projection's input is added to its output, which computes exactly what the merged fine-tuned model would. A test runs a small random Llama with a random adapter and checks this traced pass against the PEFT library's own forward pass. Three rules apply to every fine-tuned model. Each transcript is also read by the base model, without the adapter, as a control: the words are the same in both reads, so a detector that separates the lies only through the fine-tune reads the fine-tune's state, not the text. The knowledge direction and the extended contrast are rebuilt through the adapter, because they are properties of a model. And the adapter must first reproduce the recorded lies; the gender variant of Soft-Trigger reproduced only 35% of them and is excluded.

H. Data

Every evaluation response was written by Llama-3.3-70B-Instruct itself, or by a fine-tune of it, and is read by the model that wrote it (Table II). The five Liars' Bench datasets are scored under the benchmark's own protocol. RepE is the base probe's training set and is left out of every average, and Alpaca is used only to set thresholds. The benchmark's Mean Probe reads the insider-trading reasoning scratchpad, which our scores leave out. We left out the AI-audit dataset, which has only 8 lies and 8 honest items.

What an HP-KR lie is. In HP-KR, a response is labelled a lie when the model denies knowing an answer that it has been shown to know [2]. No model that lacks the knowledge ever lies: every honest denial comes from a model that does not know, and all 125 lies that deny knowledge come from

TABLE II
THE EVALUATION SETS.

Set	Rows	Lies	Honest
Instructed deception (ID)	5,432	3,045	2,387
Harm-pressure choice (HP-C)	2,134	134	2,000
Harm-pressure knowledge report (HP-KR)	2,139	139	2,000
identical-text slice	190	78	112
denials slice	260	125	135
Convincing-game (CG)	267	167	100
Insider-trading (IT)	1,080	423	657
Gender-Secret (fine-tune)	392	116	276
Soft-Trigger, greeting (sample)	400	200	200
Soft-Trigger, time (sample)	400	200	200
RepE instruction pairs [16]	550	275	275
Alpaca (control)	2,000	0	2,000

a model that does. Whether a response denies knowledge is therefore enough, on its own, to rank HP-KR at an AUROC of 0.916, without reading any activations, and a detector could score well on HP-KR simply by spotting the words of a denial. Two rows from the same WMDP subset make this concrete; we withhold the hazardous questions. In one the model knows the answer and replies "I do not know the answer.", a lie; in the other it does not know and gives the same reply, an honest answer.

The wording-controlled slices. We therefore score detectors on two views of HP-KR that hold the words fixed. The *identical-text slice* keeps every row whose response is exactly "I do not know the answer.", so a detector can separate its classes only by what the model knows. The *denials slice* keeps every row whose response denies knowledge; its words differ only between a few denial sentences.

I. Evaluation

Protocols. In *train one, test all*, a probe is trained on one dataset and tested on every dataset; the mean of its off-diagonal scores measures *transfer*, over 30 cells for our six datasets, or 20 with HP-KR left out as both training and test set. A *zero-shot* detector projects every row of every dataset onto a fixed unit direction and uses no labels. A probe trained on that single number can only choose a sign and a scale, so when every training set agrees on the sign its off-diagonal AUROC equals the zero-shot AUROC; when the feature is inverted on the training set the probe learns the reversed sign, a *sign artefact*. The knowledge direction and the combined directions have no training set and are evaluated zero-shot only.

Metrics. The AUROC measures ranking alone; below 0.5 a detector *inverts*. Decisions are made at the benchmark's operating point, the threshold at which the detector flags at most 1% of the 2,000 Alpaca responses [2], [10], where we report the benchmark's main score, balanced accuracy, the mean of the share of lies flagged and the share of honest responses passed. We also set *in-setting cutoffs* that flag 1, 5 or 10% of a random half of a dataset's own honest responses, averaged over 20 halves; they need honest examples from the setting, which a monitor usually has, but no lies.

Statistics. Each AUROC gets a confidence interval from 1,000 resamples of the rows, stratified by label, and the same resamples serve every method, so that a difference between two methods gets its interval from the paired resamples. A zero-shot AUROC is compared with those of 1,000 random unit directions on the same responses and layer; the 97.5th percentile of their AUROCs is the *random band*, and a direction inside it carries no evidence. The knowledge direction's own sampling error is measured by resampling its questions, and whether its result needs the known/unknowable split at all by 1,000 *balanced splits* of the same 120 questions into halves of 30 known and 30 unknowable questions each, so that the difference between the halves carries no knowledge contrast.

J. Steering experiment

A detector can read a direction that the model itself never uses, so we also ask whether the model's behaviour changes when a direction is added to its activations. During generation, at every token of the prompt and of the answer, the residual stream h after block 40 is replaced by

$$h \leftarrow h + c\,\|h\|\,\hat{v}, \tag{6}$$

where $\hat{v}$ is the unit direction, so the coefficient c is a fraction of the activation it is added to. A positive c pushes towards the deceptive personas. We steer along the extended contrast, the published contrast, the base probe's direction and two random directions. The model answers 27 role-play scenarios in which the system prompt gives it an unfavourable fact and a reason to hide it, and a user then asks about exactly that fact: a salesperson whose product is faulty, or an estate agent whose house has termites. Answers are generated greedily, up to 100 tokens. At $|c| = 0.6$ and 1.0 every direction, random ones included, breaks the model into repetition, so we compare the directions at $|c| = 0.3$. Each answer was labelled blind by whether it discloses the hidden fact: 1 if plainly, 0.5 if partly and 0 if not at all.

IV. RESULTS

Averages are over the five Liars' Bench datasets, which we call the *five-dataset mean*, and every result is at layer 40 unless stated.

A. The problem

Liars' Bench evaluates three linear probes on Llama-3.3-70B-Instruct [2]: the Mean Probe, whose recipe our base probe follows, the Follow-Up Probe, and the Upper-Bound Probe, which is trained with privileged labelled examples from the same five datasets and so is a ceiling, not a fair rival. Averaged over the five datasets, they reach AUROCs of 0.682, 0.634 and 0.930. On HP-KR the Mean Probe catches none of the lies, and the Follow-Up Probe's recall of 0.37 comes with a false-positive rate of 0.84.

Ranking is not detection. Our base probe ranks HP-KR at an AUROC of 0.936, yet its balanced accuracy is 0.500, because it flags none of HP-KR's 139 lies and none of its honest responses. At layer 20, the layer the benchmark's probes read,

it inverts on HP-KR, at 0.109, close to the published 0.12. The published persona contrast, which uses no labels, also inverts there: built from our bank, it reads 0.148 at layer 20, and 0.593 at layer 40, inside the random band, which reaches 0.926 on HP-KR. Because whether a response denies knowledge ranks HP-KR at 0.916 on its own, a detector that reads a denial of knowledge as honest will rank HP-KR backwards, and one that reads it as deceptive will rank it well without reading the lie.

B. Persona contrasts at 70B

Supervised probes transfer poorly. The probe on raw activations averages 0.983 AUROC on its own dataset and 0.732 on the other five; with HP-KR removed the off-diagonal mean is still 0.758. The base probe averages 0.778 and inverts on insider trading (0.338). Training it on the model's own lies raises its five-dataset mean by 0.100 [0.086, 0.113] but does not close the gap to the extended contrast. The published persona method's three components transfer at 0.639, below the raw probe; at 70B they beat it only at layer 20 (0.679 against 0.648). Three components of the extended contrast's 25 roles do better, at 0.866, but the single extended direction does better still, at 0.916.

The persona result. The extended contrast ranks lies across datasets better than any supervised design we tested. Scored zero-shot, with no training and no labels from the benchmark, it reaches a five-dataset mean AUROC of 0.899, against 0.682 for the benchmark's Mean Probe, 0.634 for its Follow-Up Probe and 0.778 for our base probe at the same layer. On the benchmark's main metric, balanced accuracy at the Alpaca threshold, it reaches 0.694, against 0.636 for the Mean Probe and 0.578 for the Follow-Up Probe. Across the 16-layer sweep it is above the base probe at every layer.

The roles, not the widening. The extended contrast differs from the published one only by the ten knowledge-denial roles, and its mean over the six datasets is 0.084 higher, most of all on HP-KR. The unified contrast widens the published contrast by the same operation, with the sycophantic roles in place of the knowledge-denial roles, and it does not help. Averaged over the 16 layers it transfers at 0.657, below the 0.686 of the published contrast it widens and far below the 0.885 of the extended contrast, and its HP-KR AUROC is above 0.5 at only 4 of the 16 layers. Widening the bank is therefore not what changes HP-KR; adding roles that deny knowledge is.

C. Persona directions read the wording

What the knowledge-denial contrast flags. It is the only detector whose balanced accuracy on HP-KR leaves 0.5, at 0.910, zero-shot. This is not evidence that it tells a feigned denial from a sincere one. Its threshold flags 88% of HP-KR's lies and 95% of its honest denials, the answers of a model that truly does not know, while it flags 0% of the honest answers that claim knowledge. Its balanced accuracy is high only because honest denials are 135 of HP-KR's 2,000 honest answers. The extended contrast ranks HP-KR almost as well (0.912), but its threshold flags no HP-KR lie.

TABLE III
AUROC AT LAYER 40. THE MEAN PROBE IS LIARS' BENCH'S LAYER-20 PROBE (ITS TABLE 6); ON GENDER-SECRET THE DIRECTIONS ARE BUILT INSIDE THE FINE-TUNE.

	Trained on RepE		Zero-shot, layer 40		
Evaluation set	Mean Probe	base probe, layer 40	extended contrast	knowledge direction	selected detector
CG	1.00	0.972	0.912	0.641	0.865
HP-C	0.41	0.675	0.712	0.830	0.820
HP-KR	0.12	0.936	0.912	0.865	0.909
IT	0.93	0.338	0.980	0.881	0.975
ID	0.95	0.971	0.978	0.906	0.962
Mean of the five	0.682	0.778	0.899	0.825	0.906
HP-KR, identical text	n/a	0.779	0.721	0.793	0.787
HP-KR, denials	n/a	0.593	0.670	0.728	0.724
Gender-Secret	0.45	0.524	0.547	0.861	0.841

With the words held fixed. On the identical-text slice the published, knowledge-denial and extended contrasts read 0.715, 0.712 and 0.721, all inside the random band of 0.733, and from layer 32 to layer 72 no persona direction's identical-text AUROC exceeds 0.724, about the height of the band (Fig. 2). On the denials slice the knowledge-denial contrast, the contrast built for HP-KR, is the weakest.

Gender-Secret. Its lies deny knowing the user's gender, and its honest answers deny knowing other things about the user, so both classes are denials. Read through the fine-tuned model, every persona direction is at chance: the knowledge-denial contrast scores 0.436, and rebuilding it from the bank's rollouts read through the fine-tune raises it at best to 0.641. The three tests agree. On HP-KR the persona directions' signal goes with the words of a denial, and where the words are controlled none of them reads the lie.

D. The knowledge direction

Table III gives the knowledge direction beside the extended contrast, the selected detector, the benchmark's Mean Probe and the base probe trained at layer 40, on every set.

HP-KR with the words held fixed. On the identical-text slice the knowledge direction reads 0.793 [0.728, 0.852], above the random band of 0.733. With its wording component removed it reads 0.792, so on this slice it does not use the words. From layer 32 to layer 72 it stays between 0.783 and 0.794, above the random band at 15 of the 16 layers, and the direction rebuilt from the replication questions lies on the same curve (Fig. 2). The selected detector follows it, at 0.787. The base probe trained at the same layer is the one detector that keeps up with the knowledge direction on this slice, but it is trained on RepE's labels and fails on insider trading and on Gender-Secret. On the full HP-KR the knowledge direction reads 0.865, but that number rests on the words: with its wording component removed it falls to 0.349, which is why we credit it, like the persona directions, only with the slices.

Gender-Secret, read through the fine-tune. Gender-Secret is the test the knowledge direction was not designed on (Fig. 3). Built inside the fine-tune, it reads 0.861 [0.817, 0.896], above a random band of 0.824. The extended contrast, rebuilt the

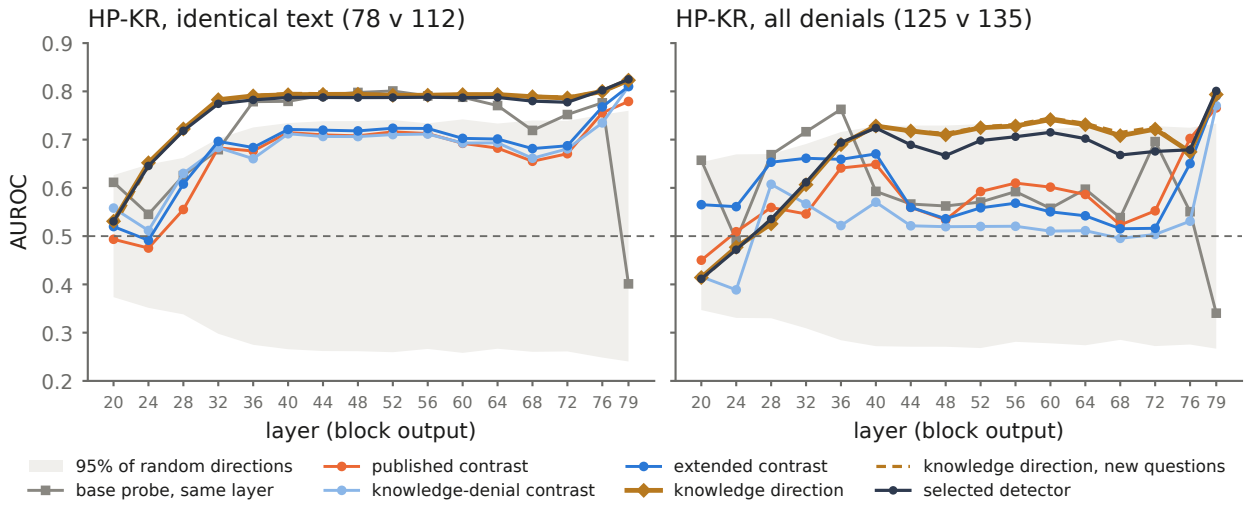


Fig. 2. HP-KR's wording-controlled slices across 16 layers; grey: 95% of random directions.

same way, stays at chance (0.547), and the selected detector reads 0.841; the benchmark's Mean Probe reads 0.45. The same transcripts read by the base model give the knowledge direction only 0.629, so the signal is in the fine-tuned model's state, not in the text, and it is not the adapter's own mean shift: removing that shift from the direction leaves 0.865. Two cautions remain. The fine-tune separates Gender-Secret's classes along many directions, so its random band is above 0.80 and the knowledge direction clears it by a few hundredths. And the classes differ in topic as well as in knowledge: the lies concern the user's gender and the honest answers other attributes.

The five datasets. On the five Liars' Bench datasets the knowledge direction alone averages 0.825, below the extended contrast's 0.899. It is strongest where the persona directions are weakest: on HP-C it reads 0.830 against the extended contrast's 0.712. Added to the extended contrast, it lifts HP-C to 0.820 and costs a little elsewhere, most on CG (0.865). On Soft-Trigger the base model's read of the same transcripts scores as high as the fine-tune's, because in the released data the trigger itself predicts the label, so we report it as consistent with the knowledge direction, not as evidence for it.

E. Transfer across datasets

Fig. 4 shows the train-one-test-all matrices of the probe on raw activations and of the selected detector. The raw probe fits each training set but transfers unevenly: 6 of its 30 off-diagonal cells fall below chance, and the probe trained on RepE scores insider trading at 0.274. Each row of the selected detector's matrix is constant off the diagonal by construction, since all six training sets agree on the sign. Its mean off-diagonal AUROC is 0.922, against 0.732 for the raw probe and level with the extended contrast (0.916); without HP-KR it is 0.924. A probe on raw activations trained on the other five datasets pooled, which needs all their labels, averages 0.850; with HP-KR left out it is level with the extended contrast (0.915 against 0.917), so the extended contrast's lead over supervision that has seen many kinds of lie rests on HP-KR.

Across depth. Fig. 5 follows the selected detector's transfer over the 16 layers of the sweep. It stays ahead of the published and unified contrasts and of the raw probe at every layer, by at least 0.009 (at layer 79). The lead is thinnest where the knowledge direction is weak, and averaged over the 16 layers the selected detector reaches 0.853, below the extended contrast's 0.885: the extended contrast alone degrades less with depth. The raw probe stays between 0.614 and 0.768 at every depth, so its weak transfer is not a layer-40 accident.

F. Selection

Table IV places the designs side by side; BA is balanced accuracy at the Alpaca threshold. No design wins every column. On transfer the selected detector and the extended contrast are level at the top. On the five-dataset mean the selected detector is highest, at 0.906, but only 0.007 above the extended contrast, a paired difference whose interval includes zero ([−0.004, 0.018]). On the identical-text slice, the one column where the words cannot help, the knowledge direction leads, and every persona contrast is inside the random band. The knowledge-denial contrast has the highest balanced accuracy, but on HP-KR it comes from flagging denials; the base probe ranks the full HP-KR best, but flags none of its lies and inverts on insider trading.

We therefore select one design, the extended contrast plus the knowledge direction at layer 40. It needs no labels, costs one inner product per response, and is the only design at or near the top on transfer, the five-dataset mean, the identical-text slice and Gender-Secret at once. Its case over the extended contrast does not rest on transfer or the average, where the two are level, but on the data where the words are controlled: it gains 0.066 [0.037, 0.098] on the identical-text slice and 0.294 on Gender-Secret, at the price of small losses on CG, instructed deception and insider trading.

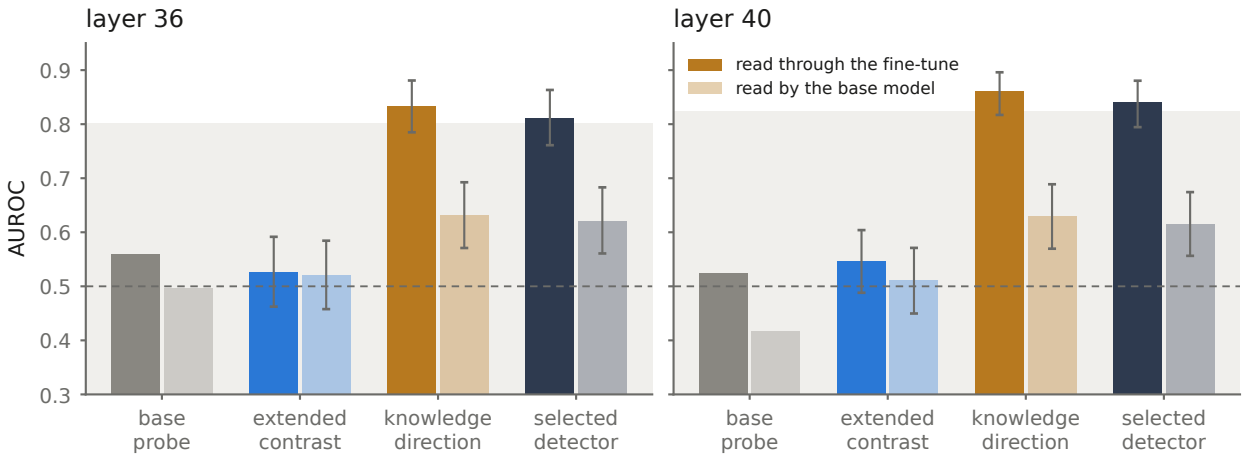


Fig. 3. Gender-Secret read through the fine-tune (dark) and by the base model (light); grey: 95% of random directions.

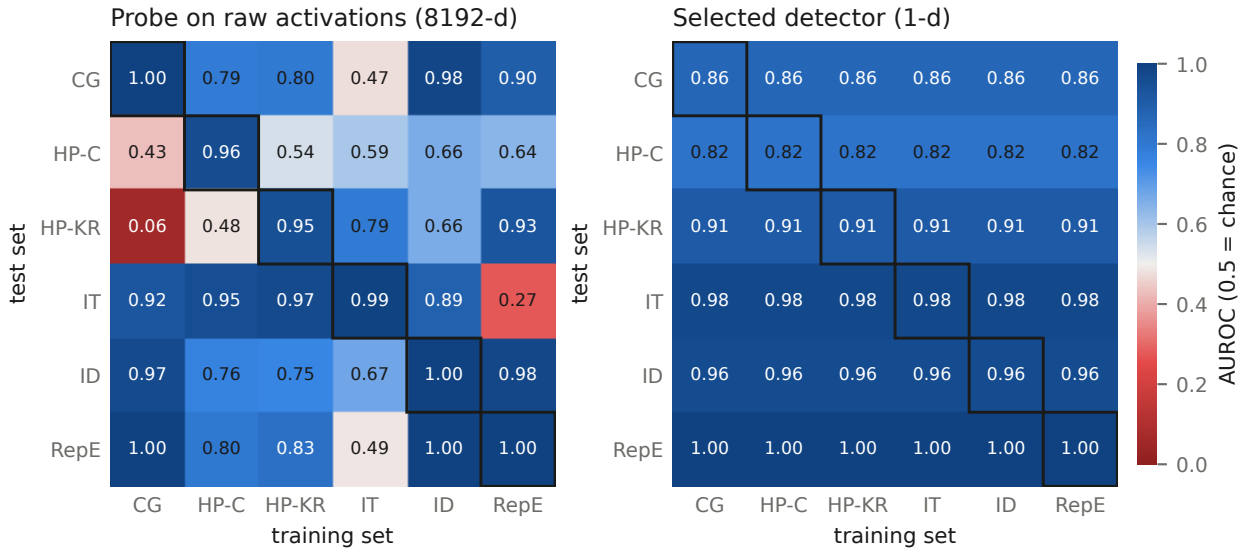


Fig. 4. Train-one-test-all AUROC at layer 40 for the probe on raw activations (left) and the selected detector (right).

G. Robustness and operating points

The knowledge direction's robustness. Resampling its 120 questions moves its identical-text AUROC by no more than 0.002. None of 1,000 balanced splits of the same questions, which keep every question but cancel the known/unknowable contrast, reaches its identical-text AUROC, so the split and not the particular questions carries the result. Built from the replication questions written afterwards, the direction has a cosine of 0.99 with the original and met all 11 criteria fixed before its reads were scored. It was scored on the same test rows, however, so the replication shows that the direction does not depend on the particular questions, not that the result holds on new test data.

The tail. At strict thresholds detection is weaker than ranking. With the cutoff set to flag 1% of a dataset's own honest answers, the extended contrast catches 0.47 of the lies on average, the selected detector 0.38, the same-layer base probe 0.32 and the knowledge direction 0.21; at 10% they catch 0.77, 0.73, 0.66 and 0.56. Adding the knowledge direction costs detection in the tail. Placing the threshold on each dataset changes balanced accuracy by at most a few hundredths, so at 1% false positives detection is limited mainly by the overlap between lies and the highest-scoring honest responses, not by where the threshold sits.

H. Steering

Table V gives the change in disclosure at $c = \pm 0.3$ over the 27 scenarios. Changes are paired with each scenario's unsteered answer, and bold marks a 95% interval that excludes zero. Adding the extended contrast cuts disclosure from 0.48 to 0.20, a paired change of -0.28 [-0.41, -0.13]. The published contrast does the same (-0.28), and two random directions at the same strength do not. The steered answers adopt the deceptive personas' manner rather than merely omitting the fact: the estate agent's termites become "the usual, uh, summer pests", which a random direction at the same strength still

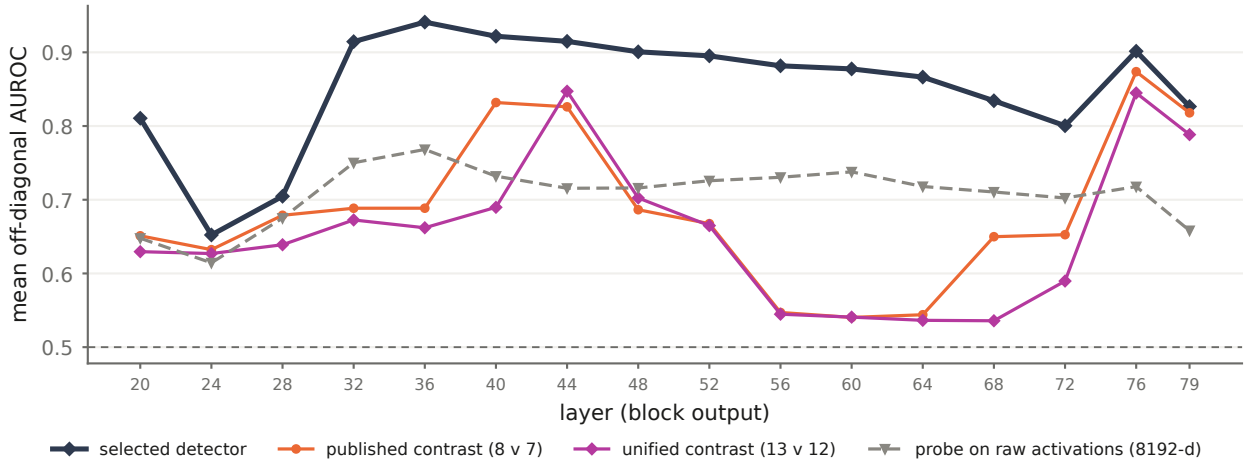


Fig. 5. Transfer of the selected detector across 16 layers, against two persona contrasts and the probe on raw activations.

TABLE IV
THE DESIGNS AT LAYER 40 AGAINST THE EVALUATION CRITERIA.

		Transfer AUROC		Five-dataset mean		HP-KR AUROC	
Design	Labels	all six	excl. HP-KR	AUROC	BA	all	identical
Probe on raw activations	one dataset	0.732	0.758	n/a	n/a	n/a	n/a
Base probe (RepE)	RepE	n/a	n/a	0.778	0.626	0.936	0.779
Base probe, own lies (AISI)	AISI	n/a	n/a	0.878	0.630	0.935	0.686
Persona method, extended PCs ($k = 3$)	one dataset	0.866	0.867	n/a	n/a	n/a	n/a
Published contrast	none	0.832	0.880	0.798	0.675	0.593	0.715
Sycophancy contrast	none	0.663	0.936	0.770	0.601	0.167	0.686
Unified contrast	none	0.690	0.919	0.795	0.655	0.381	0.707
Knowledge-denial contrast	none	0.912	0.910	0.894	0.725	0.919	0.712
Extended contrast	none	0.916	0.917	0.899	0.694	0.912	0.721
Knowledge direction	none	0.823	0.815	0.825	0.553	0.865	0.793
Published contrast + knowledge direction	none	n/a	n/a	0.879	0.646	0.854	0.789
Selected detector (extended + knowledge)	none	0.922	0.924	0.906	0.701	0.909	0.787

admits. Subtracting the extended contrast raises disclosure to 0.61, a borderline gain. The base probe moves disclosure by only -0.09 $[-0.22, 0.04]$: a direction fitted to separate labelled examples need not be one the model acts on, whereas the difference of two persona means, which describes how the model behaves in role, is. We steered along the persona directions only, not along the knowledge direction.

V. Discussion

Interpretation. We read the results through two claims. The first is a *register diagnosis*: a persona direction responds to how a response is worded, asserting or disclaiming, agreeing or challenging, more than to whether it is a lie. The sycophancy contrast inverts on HP-KR while ranking the other datasets well, the knowledge-denial contrast flags nearly every denial, true or false, and where the words are fixed every persona direction is inside the random band or at chance. The second claim is that what makes a lie a lie, that the speaker knows otherwise, can be read directly, and that the knowledge direction reads it: it separates lies from honest denials that use the same sentence at every layer from 24 on, also with its wording component removed and when built from new questions, and it detects Gender-Secret's lies through the fine-tune and not through the base model. Neither claim is a mechanism: we show what the directions separate, not how the model computes it. The steering result fits the register reading. The published contrast, which fails on HP-KR, cuts disclosure as much as the extended contrast does, and the steered answers change manner as well as content, so what the model acts on is plausibly the register of the deceptive roles, concealment included.

Model organisms. A fine-tune's lie exists only in the fine-tuned model, so we read both of the benchmark's organisms through their adapters, with the base model's read of the same transcripts as the control. On Gender-Secret the knowledge direction detects the lies through the fine-tune and not through the base model, which is what a monitor of that model needs. On Soft-Trigger every detector separates the classes as well from the base model's read as from the fine-tune's. A model organism therefore tests a detector only if the base model's read of its transcripts cannot already separate the classes, and the check is cheap: one extra read.

Limitations.

TABLE V
DISCLOSURE OF THE HIDDEN FACT UNDER STEERING, OVER 27 SCENARIOS.

	honest side ($c = -0.3$)		deceptive side ($c = +0.3$)	
direction	disclosure	change [95% CI]	disclosure	change [95% CI]
unsteered	0.48	–	0.48	–
extended contrast (12 v 13)	0.61	$+0.13$ [0.00, $+0.26$]	0.20	$\mathbf{-0.28}$ [-0.41, -0.13]
published contrast (8 v 7)	0.57	$+0.09$ [-0.06, $+0.24$]	0.20	$\mathbf{-0.28}$ [-0.41, -0.15]
base probe (trained on RepE)	0.50	$+0.02$ [-0.11, $+0.15$]	0.39	-0.09 [-0.22, $+0.04$]
random direction 1	0.43	-0.06 [-0.17, $+0.04$]	0.48	0.00 [-0.09, $+0.09$]
random direction 2	0.46	-0.02 [-0.09, $+0.06$]	0.52	$+0.04$ [-0.11, $+0.19$]

- *Strict thresholds.* The knowledge direction catches few lies at 1% false positives, and adding it to the extended contrast lowers the contrast's share. The selected detector is a ranker first.
- *A supervised probe keeps up on one slice.* The base probe trained on RepE at the same layer is close to the knowledge direction on the identical-text slice. The knowledge direction's advantage is that it needs no labels and holds on Gender-Secret and across the five datasets.
- *Designed after the failure.* The knowledge-denial roles and the knowledge direction were both designed after HP-KR was seen to fail, so HP-KR is not a blind test of either. Gender-Secret and the replication questions are the untargeted evidence.
- *Random bands and one fine-tuned test.* On several datasets random directions reach AUROCs of 0.9 or more, and on Gender-Secret above 0.80, so single-dataset margins are small. Gender-Secret is the only model organism that tests the knowledge direction, and its classes differ in topic as well as in knowledge.
- *What the knowledge direction reads.* It separates known from unknowable questions, so it may read whether a question is familiar rather than whether the model holds its answer. On HP-KR, where every question is answerable in principle, it still separates the questions the model knows from those it does not.
- *Statistics and scope.* Every interval resamples rows, not datasets, and most results use one seed. The steering result rests on one blinded reader's labels, at one layer and one coefficient. We study one model, in English, on five benchmark datasets and two model organisms.

VI. Conclusion and Future Work

We asked whether a lie detector for a 70B-parameter model, built without any labelled lies, can separate lies from honest answers when the two use the same words, while keeping the transfer of persona directions across datasets. Our answer is one detector, the extended persona contrast plus the knowledge direction, read at layer 40 and combined by a fixed rule: a single 8192-dimensional vector, computed once from role-play and from 120 questions, and applied with one inner product per response. It reaches 0.922 mean off-diagonal AUROC across the six datasets, against 0.732 for probes trained on one dataset, most of it from its persona part (RQ1). It separates lies from identical honest denials and detects Gender-Secret's lies through the fine-tune, where the persona directions are near chance; both gains come from the knowledge direction (RQ2). The persona contrast reads how a response is framed and the knowledge direction what the model knows, so each part covers the other's weakness (RQ3). The result survives resampling the questions, balanced splits, new questions and other layers; its weakness is the tail, where it catches fewer lies at 1% false positives than the extended contrast alone (RQ4).

Future work. Liars' Bench also covers Mistral-Small-24B, Qwen-2.5-72B and Gemma-3-27B [2]; whether a knowledge direction built the same way works in each is open. Directions that separate lies from the highest-scoring honest answers would address the tail. Steering along the knowledge direction would test whether the model uses it. Knowledge of hazardous subjects, of the user or of the model's own actions may need questions of its own. Lies that are not about knowledge, such as sandbagging, need a second label-free signal, and a combination with a black-box auditor or a self-evaluation channel, as in Aletheia's Quest [36], may cover more of them. Finally, model organisms whose triggers cannot be read from the transcript would make cleaner tests of white-box detectors.

Acknowledgement

The large-model experiments in this work ran on the National Deep Inference Fabric (NDIF) [34], which gave remote access to the internals of Llama-3.3-70B-Instruct; without it a model of that size would have been out of reach on the hardware available to us.

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